Edge-AI Plant Disease Classification Using Structured MobileNetV2 Compression and INT8 Quantization: A Deployment-Oriented Evaluation

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Abstract— *Plant disease classifiers can achieve high accuracy on curated leaf-image datasets, yet their storage footprint, computation, and latency often limit deployment on resource-constrained edge platforms. This paper presents a deployment-oriented MobileNetV2 pipeline that combines structured width reduction with quantization-aware training (QAT) and fully integer TensorFlow Lite inference for 38-class plant disease classification. Three structured width multipliers ($\alpha = 0.75, 0.50, 0.35$) are compared before selecting $\alpha = 0.50$ for QAT. The final INT8 artifact contains 754,902 architecture parameters and occupies 0.99 MB. In the finalized benchmark it achieves 98.03% accuracy, 2.15 ms mean latency, 2.83 ms P95 latency, and 464.55 images/s on the tested CPU runtime, a measured 45.5× mean-latency speedup over the 97.89 ms FP32 baseline. The deployment artifact is 95.37% smaller than the 21.41 MB FP32 baseline. A 24-condition corruption sweep shows that clean-test accuracy does not guarantee robustness: Gaussian blur at severity 2.0 reduces accuracy to 31.86%, while brightness, contrast, noise, and JPEG compression degrade more gracefully. These results argue for evaluating compressed edge models jointly on accuracy, compression, latency, throughput, and corruption robustness, rather than on clean-set accuracy alone.*

Index Terms— Edge AI, plant disease classification, MobileNetV2, structured compression, channel-width reduction, quantization-aware training, INT8, TensorFlow Lite, robustness evaluation.

I. INTRODUCTION

Plant diseases reduce crop yield and quality, and early, low-cost identification can help farmers intervene before a disease spreads through a field [1]. Convolutional neural networks (CNNs) have repeatedly demonstrated strong image-based plant disease recognition accuracy on curated benchmark datasets [1], but a model that performs well on such a benchmark is not automatically deployable: on-device inference for agricultural use cases (handheld scouting apps, low-cost field sensors, offline rural connectivity) requires the model to also fit tight memory and compute budgets [4].

MobileNetV2 was designed specifically around this trade-off, using inverted residual blocks and depthwise-separable convolutions to reduce the multiply-accumulate cost per layer while explicitly reporting accuracy against parameter and computation budgets [3]. This design lineage continues in MobileNetV3 [5], ShuffleNet [6], EfficientNet [7], and SqueezeNet [8], all of which target the same accuracy-versus-footprint frontier from different architectural angles. Structured compression techniques – channel pruning [9], network slimming [10], and AutoML-driven compression [11] – provide an orthogonal lever: rather than designing a smaller network from scratch, they remove channel-level capacity from an existing dense architecture so the result stays a regular, hardware-friendly model rather than an irregularly sparse one. Quantization-aware training (QAT) [12] and post-training quantization approaches such as data-free weight equalization [13] then adapt a network to low-precision arithmetic before integer-only deployment, which is what ultimately determines the on-device memory footprint and, on many CPUs and NPUs, the achievable throughput.

This work studies the complete sequence – backbone selection, structured width reduction, QAT, and INT8 export – as a single deployment pipeline, rather than isolating one compression technique. The specific contributions are:

1. A controlled comparison of three structured MobileNetV2 width multipliers ($\alpha = 0.75, 0.50, 0.35$) on a common 38-class leaf-image task, isolating the accuracy–compression trade-off before quantization is applied.
2. QAT and fully integer TensorFlow Lite deployment of the selected $\alpha = 0.50$ configuration, with an explicit weight-transfer verification step between the Keras 3 training model and the legacy `tf_keras` QAT graph.
3. A unified deployment evaluation reporting model size, parameter count, latency (mean and P95), and throughput together, rather than accuracy in isolation.
4. A 24-condition corruption robustness sweep (brightness, contrast, Gaussian noise, Gaussian blur, JPEG compression) and a class-level error analysis of the final INT8 model, which together show that a high clean-test score does not certify robustness to realistic acquisition variation.

The remainder of this paper is organized as follows. Section II reviews related work on efficient architectures, struc-

tured compression, and quantization. Section III describes the dataset, methodology, and evaluation protocol. Section IV reports the experimental results. Section V analyzes robustness and class-level errors. Section VI discusses the findings, and Section VII states threats to validity. Section VIII concludes.

II. RELATED WORK

A. Efficient CNN architectures

MobileNetV1 [4] introduced depthwise-separable convolutions to substantially cut the computation of standard convolutions with a controllable width multiplier. MobileNetV2 [3] added inverted residual blocks with linear bottlenecks, improving accuracy at a similar or lower parameter budget, and is the backbone used in this work because its width multiplier gives a simple, well-studied knob for the structured-compression comparison in Section III. MobileNetV3 [5] further tunes the architecture with neural architecture search and squeeze-and-excitation blocks; ShuffleNet [6] uses pointwise group convolutions and channel shuffling for a similar goal; EfficientNet [7] scales depth, width, and resolution jointly; and SqueezeNet [8] targets sub-megabyte model sizes directly through fire modules. This paper does not propose a new architecture in this family; it instead evaluates how far an existing, widely deployed backbone (MobileNetV2) can be pushed toward edge deployment when width reduction and quantization are combined and evaluated together.

B. Structured compression and quantization

Channel pruning [9] and network slimming [10] remove channels from a trained dense network based on learned importance, producing a smaller dense model without specialized sparse-inference kernels. AMC [11] automates the pruning ratio search per layer with reinforcement learning. The present work instead uses the width multiplier built into MobileNetV2 to obtain structured operating points directly at training time, which is simpler to reproduce but offers fewer degrees of freedom than layer-wise pruning; this trade-off is revisited in Section VI. On the quantization side, Jacob et al. [12] established the integer-arithmetic-only inference scheme used by TensorFlow Lite and the QAT formulation adopted here, while Nagel et al. [13] showed that competitive INT8 accuracy is achievable even without any fine-tuning data through weight equalization and bias correction. This paper uses QAT rather than the data-free route because labeled training data was available; a data-free comparison is noted as future work in Section VI.

C. Plant disease datasets and deployment robustness

Mohanty et al. [1] demonstrated that CNNs can reach very high accuracy on the PlantVillage-style leaf-image benchmark, but subsequent work has repeatedly flagged that such datasets are captured under controlled, uniform-background conditions that do not represent field imagery. PlantDoc [14] was built specifically to provide field-condition images after showing that models trained only on lab-style datasets transfer poorly to real photographs. Xu et al. [15] survey this gap across plant-disease datasets more broadly and argue for evaluation protocols that go beyond a single clean-test accuracy number. The present paper does not introduce a new field dataset, but it responds to the same concern from the deployment side: the 24-condition corruption sweep in Section V is designed to probe whether a model that is accurate on curated images degrades gracefully or collapses under the kind of brightness, blur, and compression variation that separates curated benchmarks from field capture.

III. METHODOLOGY

A. Dataset

The dataset used in this work contains 38,013 training images and an original test inventory of 8,146 images across 38 leaf-disease classes covering apple, blueberry, cherry, corn (maize), grape, orange, peach, pepper, potato, raspberry, soybean, squash, strawberry, and tomato. All images are 256×256 px RGB. The class taxonomy and folder-naming convention (e.g. `Apple___Apple_scab`, `Tomato___Tomato_Yellow_Leaf_Curl_Virus`) match the PlantVillage-derived benchmark introduced by Hughes and Salathé [2] and used in [1]; the dataset is treated in this paper as a PlantVillage-derived release accordingly.

B. Dataset integrity audit

An exact train/test file-path overlap check found zero overlapping paths. A subsequent SHA-256 image-hash audit identified 12 duplicate image groups in the full dataset, of which four groups crossed the training and test splits (all same-class duplicates). These four cross-split duplicate test entries were removed to obtain a leakage-free clean test set of 8,142 images; a follow-up hash check confirmed zero remaining exact train/test duplicates. This distinction explains why FP32 baseline reports in Table 1 that were computed before the audit use the original 8,146-image inventory, while all QAT INT8 and robustness results use the 8,142-image clean test set. The audit detects exact byte-level duplicates only; near-duplicate or visually similar but non-identical images are not detected, which is stated as a limitation in Section VII.

C. Backbone and structured compression

The baseline is a MobileNetV2 classifier with a 38-way softmax output and $224 \times 224 \times 3$ input, fine-tuned from ImageNet-pretrained weights. Structured reduction is controlled through MobileNetV2's width multiplier α, which scales the number of channels in every layer except the final classifier; lower α reduces parameters and FP32 model size roughly quadratically in the early layers. Three operating points were evaluated: $\alpha = 0.75, 0.50$, and 0.35. The intermediate results (Section IV-A) expose the accuracy–efficiency trade-off directly: $\alpha = 0.75$ loses substantial accuracy relative to the baseline and to $\alpha = 0.50$ in this run, while

$\alpha = 0.50$ and $\alpha = 0.35$ retain accuracy close to or above the FP32 baseline with markedly fewer parameters. $\alpha = 0.50$ was selected for QAT as the best balance between compression and clean-test accuracy among the three points tested.

D. Quantization-aware training and INT8 export

QAT was applied to the structured $\alpha = 0.50$ network for 5 epochs at a learning rate of 1×10^{-5}, monitoring validation accuracy on a held-out 10% split of the training data (34,211 train / 3,802 validation images). Because the base model was trained in Keras 3 and the QAT tooling (TensorFlow Model Optimization Toolkit, TFMOT 0.8.1) targets the legacy `tf_keras` API, backbone weights were extracted from the Keras 3 graph and transferred into an equivalent `tf_keras` architecture; a numerical verification step confirmed a maximum prediction difference of 1.22×10^{-6} between the two graphs before QAT fine-tuning began, and a parameter-count check confirmed the transferred model matched the source architecture (754,902 parameters). QAT fine-tuning improved validation accuracy from 95.24% after epoch 1 to a best of 98.26% at epoch 4 (5-epoch run, best-checkpoint restored). The fine-tuned QAT graph – which includes non-architectural quantizer variables and therefore reports 766,536 trainable parameters, versus 754,902 for the underlying architecture – was then converted to a fully integer TensorFlow Lite model using a 300-image representative dataset for calibration. The resulting artifact uses INT8 input and output tensors (input scale 0.00784, zero point -1; output scale 0.00391, zero point -128) and was verified with the TensorFlow Lite interpreter before evaluation. The final file size was 0.99 MB. Throughout the results, the parameter count reported for the QAT INT8 row is the 754,902-parameter architecture count, consistent with the FP32 rows in Table 1; the 766,536 figure that includes quantization-simulation variables is reported here for completeness but is not carried into the summary table.

E. Deployment measurements

Experiments were run with TensorFlow 2.21.0 and `tf_keras` 2.21.0 on a Windows CPU environment (no GPU acceleration; TensorFlow $\geq$2.11 does not support native Windows GPU execution). Twenty warm-up inference runs were performed before timing began. Per-image latency was measured for the full clean test set (8,142 images per condition); reported figures are the mean and 95th-percentile (P95) of these per-image latencies, excluding a small number of extreme timing outliers from the mean calculation. Throughput is computed as images processed per second of wall-clock inference time. The speedup reported is the ratio of the measured FP32 baseline mean latency to the measured QAT INT8 mean latency on this same machine, and is explicitly a platform-specific figure rather than a hardware-independent one; this caveat is repeated in Sections VI and VII because it materially limits the "edge" claim of the paper without an on-device confirmation.

Table 1: Final Deployment-Oriented Experimental Comparison

Model	α	Params	Size (MB)	Acc. (%)	Mean (ms)	P95 (ms)
Baseline FP32	1.00	2,306,662	21.41	96.24	97.89	109.33
Structured FP32	0.75	1,430,742	14.19	74.65	91.35	99.63
Structured FP32	0.50	754,902	8.95	98.20	88.12	102.79
Structured FP32	0.35	458,886	5.63	97.57	84.02	97.82
QAT INT8	0.50	754,902	0.99	98.03	2.15	2.83

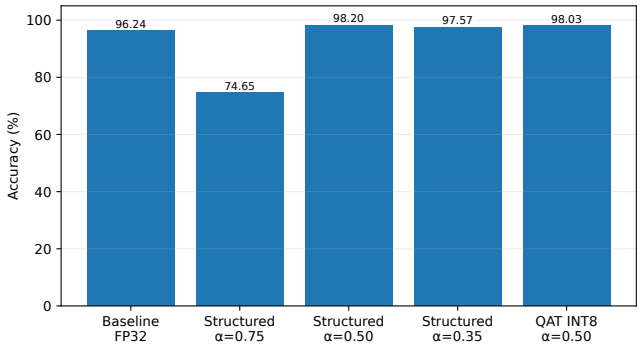


Figure 1: Accuracy comparison across the evaluated MobileNetV2 configurations.

IV. EXPERIMENTAL RESULTS

A. Model compression and clean accuracy

Table 1 summarizes the finalized deployment comparison. The FP32 baseline contains 2,306,662 parameters and occupies 21.41 MB. At $\alpha = 0.50$, structured compression reduces parameters by 67.27% and FP32 model size by 58.19% relative to the baseline, while accuracy on the same clean test protocol improves slightly, from 96.24% to 98.20%. The $\alpha = 0.75$ configuration, by contrast, drops to 74.65% accuracy in this run; the paper does not have a confirmed explanation for this non-monotonic pattern relative to $\alpha = 0.50$ and $\alpha = 0.35$; a training-instability or learning-rate-schedule cause is plausible but unverified, and is flagged as a limitation and as a direction for the repeated-run analysis recommended in Section VI.

Fig. 1 shows the same comparison graphically. The QAT INT8 model reaches 98.03% accuracy, 0.16 percentage points below the structured FP32 $\alpha = 0.50$ value of 98.20%, while providing by far the largest storage and latency reduction of any configuration tested.

B. Compression efficiency

The final QAT INT8 model reduces the deployment artifact by 95.37% relative to the FP32 baseline (21.41 MB $\rightarrow$ 0.99 MB). Fig. 2 compares parameter and model-size reduction across all configurations. Structured $\alpha = 0.35$ gives the largest parameter reduction among the FP32 variants (80.11%), but its 97.57% accuracy is below the $\alpha = 0.50$ operating point selected for QAT, so it was not carried forward. Quantization then contributes the majority of the total

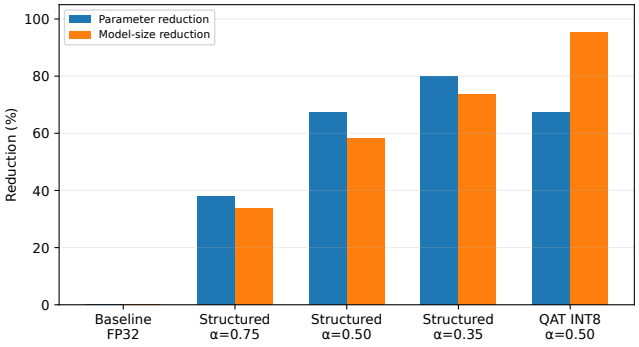


Figure 2: Parameter and model-size reductions relative to the baseline.

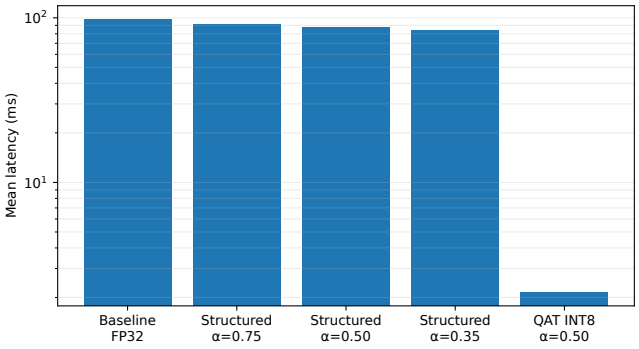


Figure 3: Mean inference latency; logarithmic vertical axis highlights the INT8 deployment gain.

size reduction on top of structured pruning: $\alpha = 0.50$ FP32 alone reduces size by 58.19%, while adding INT8 quantization brings the total reduction to 95.37%.

C. Latency and throughput

The baseline mean latency is 97.89 ms, whereas the final INT8 model records 2.15 ms, a measured 45.5× ratio. Throughput increases from 10.22 to 464.55 images/s on the same machine. As stated in Section III-E, these are CPU/TFLite runtime measurements on one Windows development machine and are reported as an internal comparison across the five configurations in Table 1, not as a claim about performance on a named embedded board; Section VII returns to this limitation. Fig. 3 uses a logarithmic latency axis because the INT8 result is separated from the FP32 measurements by more than an order of magnitude.

V. ROBUSTNESS AND ERROR ANALYSIS

A. Corruption robustness

The final QAT INT8 model was evaluated under 24 corruption conditions spanning brightness (5 levels), contrast (5 levels), Gaussian noise (5 levels), Gaussian blur (4 levels), and JPEG re-compression quality (4 levels), applied to the full 8,142-image clean test set per condition. Clean accuracy was 98.03%. Table 2 lists one representative severe setting per corruption family. Mild perturbations preserve accuracy closely (e.g. brightness scaled to 0.95, contrast scaled to 0.95, and JPEG quality 90 all stay within 0.4 percentage points of clean accuracy), but degradation becomes pronounced at severe settings and is markedly uneven across corruption types. The single most damaging condition is Gaussian blur at severity 2.0, which reduces accuracy to 31.86%, a 66.18-percentage-point drop; brightness at 0.40 and Gaussian noise at severity 20 cause comparable but smaller collapses (to 63.92% and 66.70% respectively), while JPEG compression at quality 30 – a comparatively aggressive setting – only reduces accuracy to 90.97%. Fig. 4 plots the full severity curves for all five corruption families.

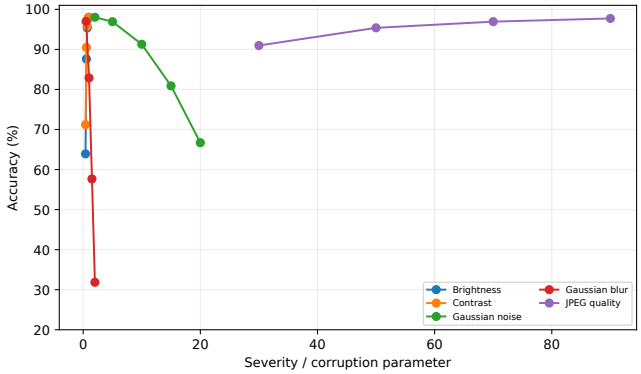


Figure 4: QAT INT8 accuracy under increasing corruption severity across five corruption families. Severity values follow the project robustness sweep protocol described in Section V-A.

Table 2: Selected Robustness Conditions (Most Severe Setting per Family)

Condition	Accuracy (%)	Drop (pp)
Clean	98.03	0.00
Brightness (scale 0.40)	63.92	34.12
Contrast (scale 0.40)	71.22	26.81
Gaussian noise ($\sigma = 20$)	66.70	31.33
Gaussian blur ($\sigma = 2.0$)	31.86	66.18
JPEG quality 30	90.97	7.06

These results are important for an edge-AI deployment claim specifically because compression and quantization say nothing, by construction, about robustness to acquisition conditions: a model can be simultaneously fast, small, and fragile. The blur sensitivity in particular is relevant to handheld or in-field photography, where camera focus, hand motion, and distance-to-leaf are not controlled the way they are in a curated dataset. This suggests that field deployment of this pipeline should be paired with image-quality gating (e.g. a blur-sharpness check before inference) or blur-augmented fine-tuning, rather than treating the clean-test accuracy number as sufficient evidence of field readiness.

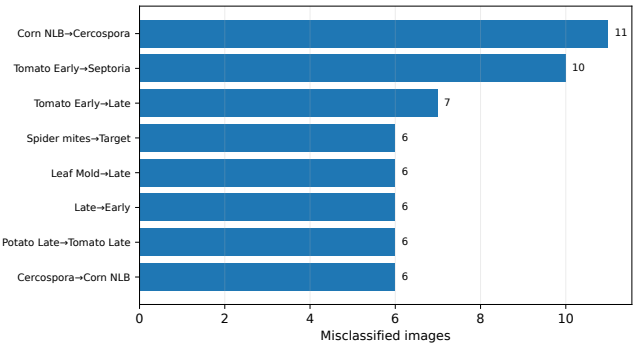


Figure 5: Largest observed QAT INT8 confusion pairs in the clean-test error analysis.

B. Class-level errors

The 38-class confusion analysis on the 8,142-image clean test set contains 7,982 correct and 160 incorrect predictions. The single largest confusion pair is Corn (maize) Northern Leaf Blight predicted as Corn Cercospora leaf spot / Gray leaf spot (11 images); other frequent confusions include Tomato Early Blight predicted as Tomato Septoria leaf spot (10 images) and Tomato Early Blight predicted as Tomato Late Blight (7 images), alongside several further tomato/potato disease pairs at 6 images each (Fig. 5). The lowest per-class recalls are Potato healthy (78.26%), Tomato Early Blight (80.67%), Tomato Leaf Mold (88.81%), Apple scab (90.43%), and Corn Cercospora leaf spot (90.91%). These errors cluster along two axes that are visually plausible confusions even for a human non-expert: (i) within-species diseases that share overlapping lesion coloration or necrotic patterning (Early Blight vs. Septoria vs. Late Blight in tomato; Northern Leaf Blight vs. Cercospora leaf spot in maize), and (ii) small-support classes such as Potato healthy (23 test images), where a handful of errors produces a large recall swing. This second point is a dataset-scale limitation rather than a modeling one, and is noted again in Section VII.

VI. DISCUSSION

The experiments show that structured compression and QAT address different, complementary parts of the deployment problem. Structured width reduction directly reduces dense-network capacity and FP32 storage; QAT then enables an integer deployment representation with a substantially smaller file and much lower measured latency, at a small (0.16 pp) additional accuracy cost relative to the structured FP32 model it was derived from. Among the three structured operating points tested, $\alpha = 0.50$ offered the best accuracy for the compression it achieves; $\alpha = 0.75$ underperformed both smaller configurations in this run, and $\alpha = 0.35$ achieved slightly lower accuracy than $\alpha = 0.50$ for a marginal additional compression gain, so it was not selected.

The results also illustrate why deployment evaluation should not be reduced to clean accuracy. The final model is highly accurate on the clean benchmark, yet severe Gaussian blur causes a large accuracy collapse, while other corruption families degrade far more gracefully. This asymmetry is only visible because accuracy, compression, latency, and robustness were measured together rather than accuracy alone being reported. It is also relevant to practical plant imaging specifically, because camera focus, hand motion, ambient lighting, and compression artifacts from message-app image sharing are all plausible sources of exactly this kind of degradation in a farmer-facing tool, even when the underlying disease taxonomy is unchanged.

Two limitations of the current comparison against the broader structured-compression and lightweight-architecture literature reviewed in Section II are worth naming directly. First, the width-multiplier approach used here offers a coarser search space than layer-wise pruning methods such as network slimming [10] or AMC [11], which could plausibly reach a similar accuracy at a smaller parameter count by pruning individual channels rather than scaling every layer uniformly; this paper does not attempt that comparison directly, and doing so is listed as future work. Second, no comparison is made against other lightweight backbones (e.g. MobileNetV3 [5], ShuffleNet [6], EfficientNet-Lite [7]) on the same dataset and evaluation protocol; the reported numbers should therefore be read as characterizing this specific MobileNetV2 pipeline, not as evidence that it is the best available option for this task.

Finally, the latency and throughput measurements are useful for comparing the five configurations in Table 1 against each other on a fixed machine, but require confirmation on a named, resource-constrained edge device (e.g. Raspberry Pi, Android phone, or microcontroller-class NPU) before they can support a hardware-specific ”edge-ready” claim; this is the single most important piece of follow-up work identified by this study.

VII. THREATS TO VALIDITY AND LIMITATIONS

First, the dataset is curated leaf imagery captured under comparatively uniform, controlled conditions, and may not represent uncontrolled field conditions (varying backgrounds, lighting, occlusion, multiple leaves per frame); this is the reason the robustness sweep in Section V was performed, but a synthetic corruption sweep is not a substitute for evaluation on genuinely field-captured images such as PlantDoc [14]. Second, the leakage audit removes exact cross-split byte-level duplicates but does not prove that all visually near-duplicate or same-leaf images are separated between splits. Third, the final QAT clean-test evaluation contains 8,142 images after duplicate removal, whereas some FP32 baseline figures in Table 1 that were computed before the audit use the original 8,146-image inventory; this distinction is reported explicitly rather than treating the two counts as interchangeable, and it introduces a small, bounded inconsistency across rows of Table 1 that a reader comparing FP32 and INT8 rows precisely should be aware of. Fourth, latency and throughput are runtime-specific measurements from one tested Windows CPU/TFLite environment and have not yet been confirmed

on a named edge device, as discussed in Section VI. Fifth, the robustness sweep uses five synthetic corruption families at a finite set of severities and cannot represent every real-world acquisition failure mode (e.g. partial occlusion, multi-leaf frames, non-leaf backgrounds). Sixth, results in Table 1 and 2 are reported from a single training/evaluation run per configuration rather than averaged over multiple seeds; the reported figures should be read as point estimates rather than as having a characterized run-to-run variance, and the four-decimal precision used internally during experimentation is rounded to two decimals throughout this paper to avoid implying more precision than a single run supports. Finally, the study evaluates image-level disease classification rather than disease localization, severity estimation, or multi-disease detection within a single image.

VIII. CONCLUSION

This paper presented a deployment-oriented plant disease classification pipeline based on MobileNetV2, structured width reduction, and QAT INT8 quantization, evaluated jointly on accuracy, compression, latency, throughput, and corruption robustness rather than on accuracy alone. Across three structured operating points, $\alpha = 0.50$ offered the best balance for subsequent quantization in this experiment. The final QAT INT8 artifact occupies 0.99 MB, achieves 98.03% accuracy, records 2.15 ms mean latency and 2.83 ms P95 latency, and reaches 464.55 images/s in the tested CPU runtime. Relative to the FP32 baseline artifact, the final deployment representation is 95.37% smaller, with a measured 45.5$\times$ mean-latency ratio on the same machine. The robustness sweep provides an important qualification to these efficiency numbers: severe Gaussian blur can reduce accuracy to 31.86%, even though other corruption types degrade far more gracefully. The central result of this study is therefore not simply a high classification score or a large compression ratio in isolation, but a compact deployment configuration reported alongside explicit efficiency and robustness measurements, together with an honest accounting of where the current evaluation still falls short of an on-device, field-validated "edge-AI" claim. Future work should (i) benchmark the INT8 artifact on a named ARM/embedded platform and report memory, energy, and thermal behavior in addition to latency; (ii) evaluate on field-captured imagery such as PlantDoc [14] rather than curated images alone; (iii) compare the width-multiplier compression path used here against layer-wise structured pruning [10, 11] and against other lightweight backbones [5–7] under an identical protocol; and (iv) repeat the training and evaluation pipeline over multiple random seeds to report variance alongside the point estimates given here.

REPRODUCIBILITY

The dataset splits, structured $\alpha = 0.50$ Keras model, QAT INT8 TensorFlow Lite artifact, dataset integrity/duplicate-audit reports, per-class classification reports, and the full 24-condition robustness sweep results referenced in this paper are retained as project artifacts and are available from the author on request.

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